PHYS-1320
EXAM ONE
Given Time: 2:00 hrs.
March 23, 2026

Name: _____ Student ID: _____

Breakout Session _____

Relevant Equations and Constants

Coulomb constant $k_e = \frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9$ N/m^2/C^2

Electron mass 9.11×10^{-31} kg = 5.485×10^{-4} u

Electron Charge -1.602×10^{-19} C

Permittivity of free space 8.854×10^{-12} C^2/Nm^2

Conversion between Joule and Electron Volt

Gauss's Law differential and Integral Form:

$$\vec{\nabla} \cdot \vec{\mathbf{E}} = \frac{1}{\epsilon_0}\rho$$

Gauss's law

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{a}} = \frac{1}{\epsilon_o} Q_{enc}$$

Coulombs Law $F_e = k_e \frac{|q_1||q_2|}{r^2}$

Electric Potential $\Delta V \equiv \frac{\Delta U}{q} = -\int_{Ⓐ}^{Ⓑ} \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}}$

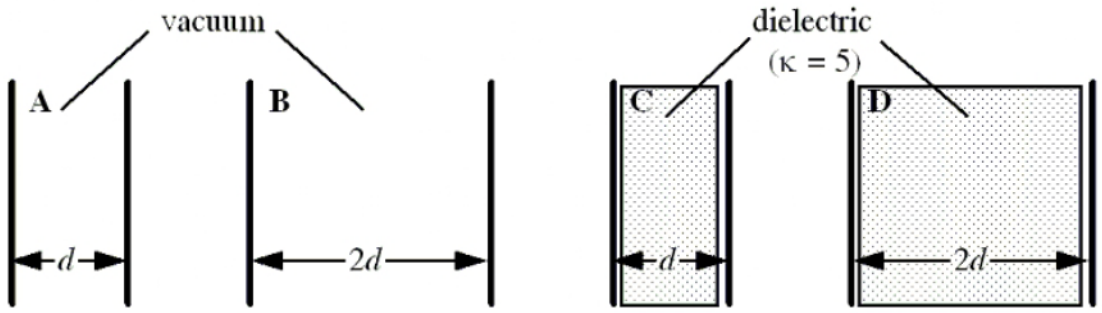
Divergence

$$\nabla \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

Conversion between Joule (J) and electron Volt

1eV=1.602 x10-^{19}J

The figure below shows four parallel plate capacitors: **A**, **B**, **C**, and **D**. Each capacitor carries the same charge q and has the same plate area A. As suggested by the figure, the plates of capacitors **A** and **C** are separated by a distance d while those of **B** and **D** are separated by a distance $2d$. Capacitors **A** and **B** are maintained in vacuum while capacitors **C** and **D** contain dielectrics with constant $\kappa = 5$.



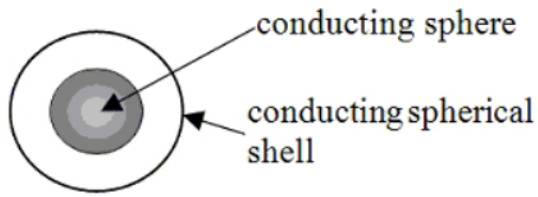
1. Which list below places the capacitors in order of *increasing* capacitance?
A) **A**, **B**, **C**, **D**
B) **B**, **A**, **C**, **D**
C) **A**, **B**, **D**, **C**
D) **B**, **A**, **D**, **C**
E) **D**, **C**, **B**, **A**

2. Which capacitor has the largest potential difference between its plates?
A) **A**
B) **B**
C) **C**
D) **D**
E) **A** and **D** are the same and larger than **B** or **C**.

3 . Which capacitor is storing the greatest amount of electric potential energy?
A) **A**
B) **B**
C) **C**
D) **D**
E) Since all four carry the same charge, each will store the same amount of energy.

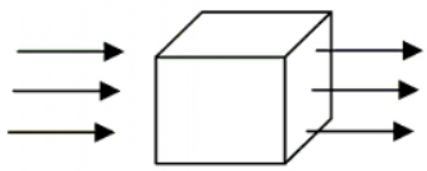
4. The magnitude of the electric field at a distance of two meters from a negative point charge is E. What is the magnitude of the electric field at the same location if the magnitude of the charge is doubled?
A) $E/4$ B) $E/2$ C) E D) $2E$ E) $4E$

5. A conducting sphere carries a net charge of +6 μC. The sphere is located at the center of a conducting spherical shell that carries a net charge of –2 μC. Determine the excess charge on the outer surface of the spherical shell.



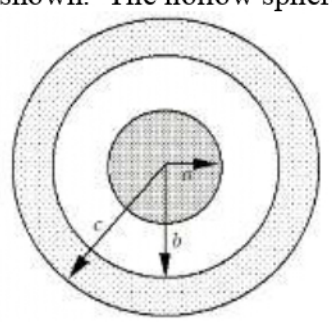
A) –4 μC
B) +4 μC
C) –8 μC
D) +8 μC
E) +6 μC

6. A cubical Gaussian surface is placed in a uniform electric field as shown in the figure. The length of each edge of the cube is 1.0 m. The uniform electric field has a magnitude of 5.0×10^8 N/C and passes through the left and right sides of the cube perpendicular to the surface. What is the total electric flux that passes through the cubical Gaussian surface?



A) 5.0×10^8 N·m²/C
B) 3.0×10^9 N·m²/C
C) 2.5×10^6 N·m²/C
D) 1.5×10^7 N·m²/C
E) zero N·m²/C

For Question 7 and 8
A solid, conducting sphere of radius *a* carry an excess charge of +6 μC. This sphere is located at the center of a hollow, conducting sphere with an inner radius of *b* and an outer radius of *c* as shown. The hollow sphere also carries a total excess charge of +6 μC.



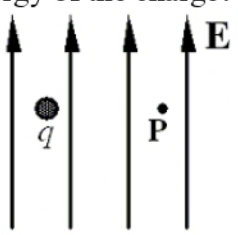
7. Determine the excess charge on the *inner surface* of the outer sphere (a distance *b* from the center of the system).
A) zero coulombs B) –6 μC C) +6 μC D) +12 μC E) –12 μC

8. Determine the excess charge on the *outer surface* of the outer sphere (a distance *c* from the center of the system).
A) zero coulombs B) –6 μC C) +6 μC D) +12 μC E) –12 μC

9. Which one of the following statements best explains why it is possible to define an *electrostatic potential* in a region of space that contains an *electrostatic field*?
A) Work must be done to bring two positive charges closer together.
B) Like charges repel one another and unlike charges attract one another.
C) A positive charge will gain kinetic energy as it approaches a negative charge.
D) The work required to bring two charges together is independent of the path taken.
E) A negative charge will gain kinetic energy as it moves away from another negative charge.

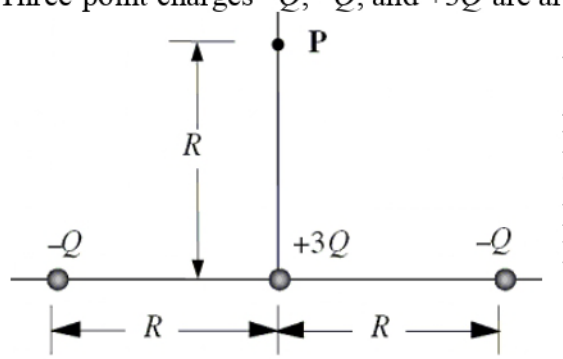
10. Two positive point charges are separated by a distance R. If the distance between the charges is reduced to $R/2$, what happens to the total electric potential energy of the system?
A) The total electric potential energy is doubled.
B) The total electric potential energy remains the same.
C) The total electric potential energy increases by a factor of 4.
D) The total electric potential energy is reduced to one-half of its original value.
E) The total electric potential energy is reduced to one-fourth of its original value.

11. A charge $q = -6.0\ \mu C$ is moved 0.25 m horizontally to point **P** in a region where an electric field is 250 V/m directed vertically, as shown. What is the change in the electric potential energy of the charge?



A) -2.4×10^{-5} J
B) -1.5×10^{-4} J
C) zero joules
D) $+1.5 \times 10^{-4}$ J
E) $+2.4 \times 10^{-5}$ J

12. Three-point charges $-Q$, $-Q$, and $+3Q$ are arranged along a line as shown in the sketch.



What is the electric potential at the point **P**?
A) $+kQ/R$
B) $-2kQ/R$
C) $-1.6kQ/R$
D) $+1.6kQ/R$
E) $+4.4kQ/R$

13. If the work required to move a +0.25 C charge from point **A** to point **B** is +175 J, what is the potential difference between the two points?
A) zero volts B) 44 V C) 88 V D) 350 V E) 700 V

An isolated system consists of two conducting spheres **A** and **B**. Sphere **A** has five times the radius of sphere **B**. Initially, the spheres are given equal amounts of positive charge and are isolated from each other. The two spheres are then connected by a conducting wire.
Note: The potential of a sphere of radius R that carries a charge Q is $V = kQ/R$, if the potential at infinity is zero.

14. Which one of the following statements is true *after* the spheres are connected by the wire?
A) The electric potential of **A** is 1/25 as large as that of **B**.
B) The electric potential of **A** equals that of **B**.
C) The electric potential of **A** is 25 times larger than that of **B**.
D) The electric potential of **A** is 1/5 as large as that of **B**.
E) The electric potential of **A** is five times larger than that of **B**.

15. Determine the ratio of the charge on sphere **A** to that on sphere **B**, q_A/q_B, after the spheres are connected by the wire.
A) 1 B) 1/5 C) 5 D) 2 E) 1/25

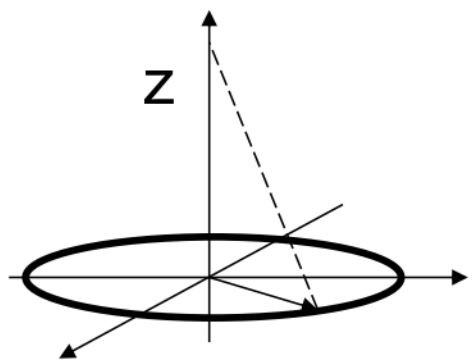
16. A uniformly charged ring, in the xy plane, centered on the origin, has radius a and total charge Q. V(r = ∞) = 0. What is the voltage at z on the z-axis?

A. $\frac{kQ}{a}$

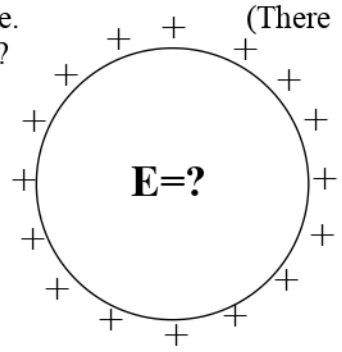
B. $\frac{kQ}{z}$

C. $\frac{kQ}{\sqrt{a^2+z^2}}$

D. $\frac{kQ}{a^2+z^2}$



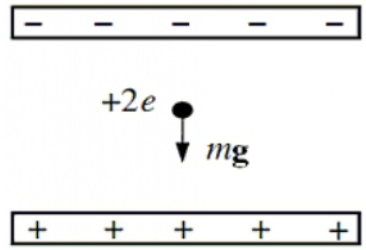
17. A spherical *shell* has a uniform positive charge density on its surface. (There are no other charges around) What is the electric field *inside* the sphere?
A: **E**=0 everywhere inside
B: **E** is non-zero everywhere in the sphere
C: **E**=0 only at the very center, but non-zero elsewhere inside the sphere.
D: Not enough info given



18. Given a spherical SHELL with uniform *surface charge* density σ (no other charges anywhere else) what can you say about the potential V *inside* this sphere? (Assume as usual, V(∞)=0)
A) V = 0 everywhere inside
B) V = non-zero constant everywhere inside
C) V must vary with position but is zero at the center.
D) Could be A OR B!
E) None of these, it's something else!

19. A total charge of –6.50 μC is uniformly distributed within a sphere that has a radius of 0.150 m. What is the magnitude and direction of the electric field at 0.300 m from the surface of the sphere?
A) 2.89×10^5 N/C, radially inward
B) 6.49×10^5 N/C, radially outward
C) 4.69×10^5 N/C, radially inward
D) 9.38×10^5 N/C, radially outward
E) 1.30×10^6 N/C, radially inward

20. A helium nucleus is located between the plates of a parallel-plate capacitor as shown. The nucleus has a charge of +2e and a mass of 6.6×10^{-27} kg. What is the magnitude of the electric field such that the electric force exactly balances the weight of the helium nucleus so that it remains stationary?

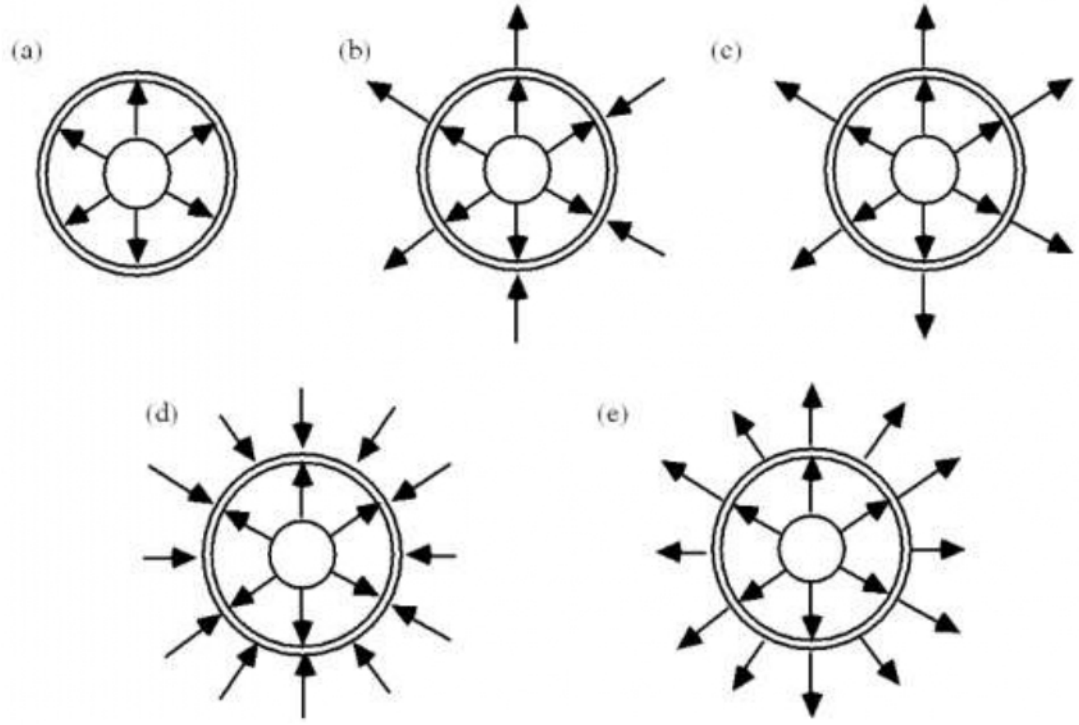


A) 4.0×10^{-7} N/C
B) 6.6×10^{-26} N/C
C) 2.0×10^{-7} N/C
D) 5.0×10^{-3} N/C
E) 1.4×10^{-8} N/C

21. Two identical conducting spheres carry charges of +5.0 μC and –1.0 μC, respectively. The centers of the spheres are initially separated by a distance *L*. The two spheres are brought together so that they are in contact. The spheres are then returned to their original separation *L*. What is the ratio of the magnitude of the electric force on either sphere *after* the spheres are touched to that *before* they were touched?

A) 1/1 B) 4/5 C) 9/5 D) 5/1 E) 4/9

22. Which one of the following figures shows a qualitatively accurate sketch of the electric field lines in and around this system?



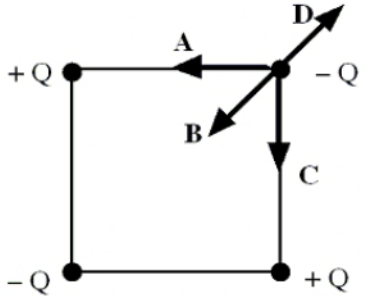
A) a B) b C) c D) d E) e

23. An aluminum nail has an excess charge of +3.2 μC. How many electrons must be added to

the nail to make it electrically neutral?
A) 2.0×10^{13} B) 2.0×10^{19} C) 3.2×10^{16} D) 3.2×10^{6}
E) 5.0×10^{-14}

24. Two charged particles **A** and **B** are located near one another. Both the *magnitude* and *direction* of the force that particle **A** exerts on particle **B** is *independent* of
A) the sign of charge **B**.
B) the sign of charge **A**.
C) the distance between **A** and **B**.
D) the magnitude of the charge on **B**.
E) The magnitude and direction of the force are dependent on all the above choices.

25. Four-point charges, each of the same magnitude, with varying signs are arranged at the corners of a square as shown. Which of the arrows labeled **A**, **B**, **C**, and **D** gives the correct direction of the net force that acts on the charge at the upper right corner?

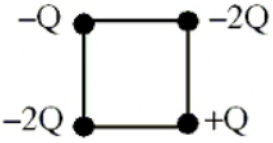


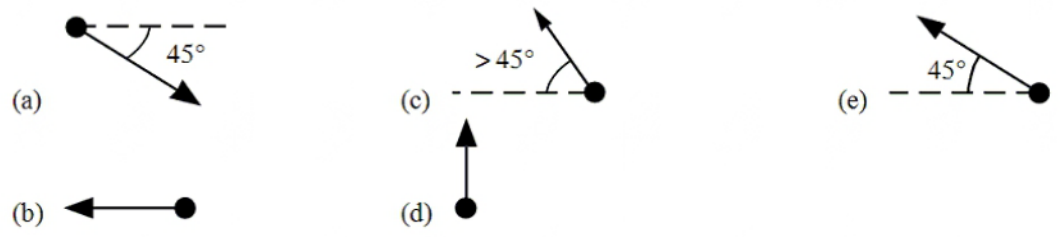
A) **A**
B) **B**
C) **C**
D) **D**
E) The net force on that charge is zero.

26. Two positive point charges Q and $2Q$ are separated by a distance R. If the charge Q experiences a force of magnitude F when the separation is R, what is the magnitude of the force on the charge $2Q$ when the separation is $2R$?
A) $F/4$ B) $F/2$ C) F D) $2F$ E) $4F$

27. A charge Q exerts a 1.2 N force on another charge q. If the distance between the charges is doubled, what is the magnitude of the force exerted on Q by q?
A) 0.30 N B) 0.60 N C) 2.4 N D) 3.6 N E) 4.8 N

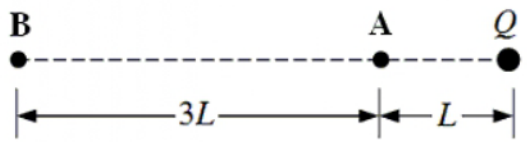
28. Four-point charges are held fixed at the corners of a square as shown in the figure. Which of the five arrows shown below most accurately shows the direction of the net force on the charge –Q due to the presence of the three other charges?





A) a B) b C) c D) d E) e

29. In the figure, point **A** is a distance L away from a point charge Q. Point **B** is a distance $4L$ away from Q. What is the ratio of the electric field at **B** to that at **A**, E_B/E_A?



A) 1/16 B) 1/9 C) ¼ D) 1/3
E) This cannot be determined since neither the value of Q nor the length L is specified.

30. A free electron and a free proton are released in identical electric fields. How do the magnitudes of the electric force exerted on the two particles compare?

(a) It is millions of times greater for the electron.
(b) It is thousands of times greater for the electron.
(c) They are equal.
(d) It is thousands of times smaller for the electron.
(e) It is millions of times smaller for the electron.

31. What prevents gravity from pulling you through the ground to the center of the Earth? Choose the best answer.

(a) The density of matter is too great.
(b) The positive nuclei of your body's atoms repel the positive nuclei of the atoms of the ground.
(c) The density of the ground is greater than the density of your body.
(d) Atoms are bound together by chemical bonds.
(e) Electrons on the ground's surface and the surface of your feet repel one another.

32. A very small ball has a mass of 5.00×10^{-3} kg and a charge of 4.00 μC. What magnitude electric field directed upward will balance the weight of the ball so that the ball is suspended motionless above the ground?

(a) 8.21×10^{2} N/C (b) 1.22×10^{4} N/C (c) 2.00×10^{-2} N/C
(d) 5.11×10^{6} N/C (e) 3.72×10^{3} N/C

33. An electron with a speed of 3.00×10^6 m/s moves into a uniform electric field of magnitude 1.00×10^3 N/C. The field lines are parallel to the electron's velocity and pointing in the same direction as the velocity. How far does the electron travel before it is brought to rest?

(a) 2.56 cm (b) 5.12 cm (c) 11.2 cm (d) 3.34 m (e) 4.24 m

34. A point charge of -4.00 nC is located at (0, 1.00) m. What is the x component of the electric field due to the point charge at (4.00, -2.00) m?

(a) 1.15 N/C (b) 20.864 N/C (c)1.44 N/C (d) -1.15 N/C
(e) 0.864 N/C

35. A circular ring of charge with radius b has total charge q uniformly distributed around it. What is the magnitude of the electric field at the center of the ring?

(a) 0 (b) $k_e q/b^2$ (c) $k_e q^2/b^2$ (d) $k_e q^2/b$ (e) none of those answers

36. What happens when a charged insulator is placed near an uncharged metallic object?

(a) They repel each other.
(b) They attract each other.
(c) They may attract or repel each other, depending on whether the charge on the insulator is positive or negative.
(d) They exert no electrostatic force on each other.
(e) The charged insulator always spontaneously discharges.

37. Estimate the magnitude of the electric field due to the proton in a hydrogen atom at 5.29×10^{-11} m, the expected position of the electron in the atom.

(a) 10^{-11} N/C (b) 10^8 N/C (c) 10^{14} N/C (d) 10^6 N/C
(e) 10^{12} N/C

38. A cubical gaussian surface surrounds a long, straight, charged filament that passes perpendicularly through two opposite faces. No other charges are nearby. Over how many of the cube's faces is the electric field zero?

(a) 0 (b) 2 (c) 4 (d) 6

39. A coaxial cable consists of a long, straight filament surrounded by a long, coaxial, cylindrical conducting shell. Assume charge Q is on the filament, zero net charge is on the shell, and the electric field is E_1 at a particular point P *(lies along the x axis)* midway between the filament and the inner surface of the shell. Next, you place the cable into a uniform external field $2E_1$ What is the x com- ponent of the electric field at P then?

(a) 0 b) between 0 and E_1 (c) E_1 (d) between 0 and $2E_1$ e) $2E_1$

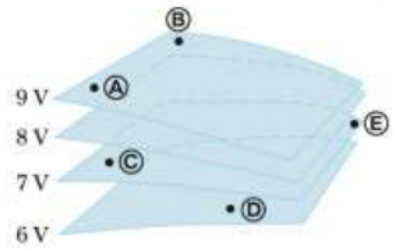
40. In which of the following contexts can Gauss's law *not* be readily applied to find the electric field?

(a) near a long, uniformly charged wire
(b) above a large, uniformly charged plane
(c) inside a uniformly charged ball
(d) outside a uniformly charged sphere
(e) Gauss's law can be readily applied to find the electric field in all these contexts.

41. A particle with charge q is located inside a cubical gaussian surface. No other charges are nearby. If the particle is at the center of the cube, what is the flux through each one of the faces of the cube?

(a) 0 (b) $q/2\varepsilon_0$ (c) $q/6\varepsilon_0$ (d) $q/8\varepsilon_0$
(e) depends on the size of the cube

42. Consider the equipotential surfaces shown in Figure. In this region of space, what is the approximate direction of the electric field?



(a) It is out of the page.
(b) It is into the page.
(c) It is toward the top of the page.
(d) It is toward the bottom of the page.
(e) The field is zero.

43. A metallic sphere A of radius 1.00 cm is several centimeters away from a metallic spherical shell B of radius 2.00 cm. Charge 450 nC is placed on A, with no charge on B or anywhere nearby. Next, the two objects are joined by a long, thin, metallic wire (as shown in Fig.), and finally the wire is removed. How is the charge shared between A and B?

(a) 0 on A, 450 nC on B
(b) 90.0 nC on A and 360 nC on B, with equal surface charge densities
(c) 150 nC on A and 300 nC on B
(d) 225 nC on A and 225 nC on B
(e) 450 nC on A and 0 on B

44. _The electric potential at x =3.00 m is 120 V, and the electric potential at x = 5.00 m is 190 V. What is the x component of the electric field in this region, assuming the field is uniform?

(a) 140 N/C (b) -140 N/C (c) 35.0 N/C (d) -35.0 N/C
(e) 75.0 N/C

45. In a certain region of space, a uniform electric field is in the x direction. A particle with negative charge is carried from $x = 20.0$ cm to $x = 60.0$ cm. Does the electric potential energy of the charge–field system

(a) increase, (b) remain constant, (c) decrease, or
(d) change unpredictably?

46. An electron in an x-ray machine is accelerated through a potential difference of 1.00×10^4 V before it hits the target. What is the kinetic energy of the electron in electron volts?

(a) 1.00×10^4 eV (b) 1.60×10^{-15} eV (c) 1.60×10^{-22} eV
(d) 6.25×10^{22} eV (e) 1.60×10^{-19} eV

47. By what factor is the capacitance of a metal sphere multiplied if its volume is tripled?

(a) 3 (b) $3^{1/3}$ (c) 1 (d) $3^{-1/3}$ (e) $^{1/3}$

48. An electronics technician wishes to construct a parallel-plate capacitor using rutile (k 5 100) as the dielectric. The area of the plates is 1.00 cm^2. What is the capacitance if the rutile thickness is 1.00 mm?

(a) 88.5 pF (b) 177 pF (c) 8.85 μF (d) 100 μF (e) 35.4 μF

49. A parallel-plate capacitor is connected to a battery. What happens to the stored energy if the plate separation is doubled while the capacitor remains connected to the battery?

(a) It remains the same. (b) It is doubled. (c) It decreases by a factor of 2.
(d) It decreases by a factor of 4. (e) It increases by a factor of 4.

50. If three unequal capacitors, initially uncharged, are connected in series across a battery, which of the following statements is true?
(a) The equivalent capacitance is greater than any of the individual capacitances.
(b) The largest voltage appears across the smallest capacitance.
(c) The largest voltage appears across the largest capacitance.
(d) The capacitor with the largest capacitance has the greatest charge.
(e) The capacitor with the smallest capacitance has the smallest charge.

EXTRA CREDIT- 5-point problem

51. What is the charge density of a field described by $E = r\hat{\boldsymbol{r}}$
a) $3\varepsilon_o$
b) ε_o
c) $q/r\pi r^2$
d) 0

PHYS-1320
EXAM ONE: Answer Sheet

Name: _____ Student ID: _____

Breakout Session _____

Multiple Choice

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47. _____
48. _____
49. _____
50. _____

Bonus qxn. worth 5 pts

51. _____